WiFi CSI People-Counting and Presence Datasets

ESP32 hardware: WiFlow (ESP32-WROOM-32U), WiCount (ESP32-S3), WiFiVision (ESP32, chip not specified), OfficeLocalization (ESP32-C6), Less is More (ESP32-C3), IIT (BHU) (ESP32 NodeMCU), Haron98 (ESP32, model not stated) and the OTH thesis set (ESP32-S3). The other five use Intel IWL-5300, Broadcom/Nexmon or Atheros cards. **[Note]** marks a point where the original dataset notes don't match the sources.

Comparison

#	Dataset	Hardware	Collection	Features / format	Size	Max count	Missing counts / imbalance	Still vs moving
1	RadioPoints WiFi-CrowdCounting	Intel IWL-5300	Modified IWL-5300 driver (Halperin CSI Tool). Band and bandwidth not specified. ~43-54 pkt/s (measured)	Raw .dat files read with read_bf_file.m: complex CSI, 3 RX x 1-2 TX antennas (measured)	26 recordings, ~91 MB total. ~5.7k-10k packets each, ~2-3 min (5 files parsed)	**8** (rooms A and C)	One recording per count per room. Room B has no 8p	Could not verify
2	WiFlow (paper)	2x ESP32-WROOM-32U with external antennas; Raspberry Pi logger	Hernandez-Bulut ESP32 CSI tool, ~100 Hz. Band not specified	52 subcarriers. File format not specified	Not specified; paper calls it "proprietary, in-house"	9	Groups of 2, 5 and 9 only. Labels are enter/exit/no-event, not headcounts	Moving (people entering and leaving)
3	EHUNAM (figshare)	MC1 and MC3: Raspberry Pi 4, BCM43455C0. MC2: Atheros QCWB335 TX / AR5822 RX	Nexmon CSI (MC1, MC3) and Atheros CSI Tool (MC2). 5 GHz ch40 at 20/80 MHz; 2.4 GHz ch13 at 20 MHz	.mat files: complex CSI, RSSI, timestamps, metadata. 56, 208 or 241 usable subcarriers	2,401 files of up to 60 s, ~38 h. Size not specified	**8** (not 14)	Big gap from 4 to 5 (see breakdown)	Moving: people walked freely
4	80 MHz Human Sensing (paper)	TX/RX: Netgear X4S AC2600, Asus RT-AC86U, TP-Link AD7200. Monitor: Asus RT-AC86U, 4 antennas	Nexmon CSI. 5 GHz ch42, 80 MHz, 173 pkt/s	.mat complex CFR, (N x 4) x 242 subcarriers	>13 h, 23.6 GB, 26 folders (4 for people counting, PC)	10	**n01 missing** from the counting folders (1-person traces are in the separate person-identification folders, PI). Per-count trace counts: Could not verify (paywalled)	Moving
5	WiCount	ESP32-S3, 1 TX + multiple RX, 1 antenna each	2.4 GHz, 20 MHz. ~100 Hz claimed, 80-94 Hz measured	CSV: 52 complex subcarriers (104 ints), RSSI, timestamps, ESP32 metadata	15 x 60 s recordings, 78,618 packets, ~49 MB	3	0:1, 1:7, 2:6, 3:1 (severe at both ends)	Not specified
6	WiMANS (paper)	2x HP EliteDesk 800 G2, Intel 5300, 3 antennas each	Linux 802.11n CSI Tool. 2.4 GHz ch12 and 5 GHz ch64, 1000 pkt/s	Raw .mat, amplitude .npy, video, annotation.csv. 30 subcarriers x 3x3 antennas	11,286 x 3 s samples, 9.4 h. Size: Could not verify	5	0: 594 (5.3%), 1: 3,564, 2-5: 1,782 each	Both (activity labels)
7	CountFi (paper)	2 pingers + 2 Raspberry Pi receivers, single antenna. Pi model and chip not specified	Nexmon CSI + tcpdump, decoded with Nexcsi. Band and bandwidth not specified	PCAP files, 256 subcarriers (242 used), plus RSSI	>19.96 M packets. Paper: 16,590 derived segments	20	21 classes (0-20) in the paper. Recordings per class: Could not verify	Mostly still (seated); April 2025 set mixed
8	WiFiVision Counting-Tools	2x "ESP32-CP2102" boards (exact chip not specified), 1 TX + 1 RX, 11 m apart. Antennas not specified	CSI firmware not public ("contact for ESP32 code"). Band, bandwidth and rate not specified	csi/ files + camera frames + labels.csv / groundtruth.csv. CSI columns not specified	9,600 samples (8,000 train / 1,600 test), 8 CSI files per split, ~2 GB. On request only	7	1,200 samples per class (0-7), balanced. Samples are not independent recordings; recordings per count: Not specified	Not specified
9	OfficeLocalization	2x ESP32-C6 (TX in AP mode, RX in STA mode), ~3 m LOS	802.11n HT-LTF, 20 MHz, ~200 pkt/s (irregular). Channel not specified	CSV per packet: 128 ints (64 complex subcarriers), RSSI, noise floor, AGC/FFT gain, channel, local timestamp	11 CSV files on Hugging Face (empty_1-5, one_1-2, two_1-2, five_1-2), ~0.95 GB	1	**Not a counting dataset**: labels are empty / person in zone 1, 2 or 5	Not specified
10	Less is More mesh	9-node ESP32-C3 mesh, 72 directional links. Antennas not specified	~10 Hz per link. Tool, band and bandwidth not specified	52 complex subcarriers per sample (CSI archives); hourly label CSV, YAML diary, door-sensor and camera-metadata files	>300 h continuous (312 labelled hours), 46.0 GB, ~1.6 M windows; one apartment, 13 days	2 (household)	**No per-hour headcount**: hourly labels are empty 99 h / sleep 104 h / active 109 h	Both, as activity tags (e.g. evening_stationary, morning_movement)
11	IIT (BHU) multi-human	ESP32 NodeMCU, TX-RX pair. Antennas not specified	ESP32-CSI Toolkit. Band, bandwidth and rate not specified	Single 270.84 MB ZIP; contents not documented on the page (paywalled)	270.84 MB. File and recording counts: Could not verify	3	1-, 2- and 3-user scenarios. **0 people: not documented**. Per-count recordings: Could not verify	Moving (6 activities); this is an activity-recognition dataset

#	Dataset	Hardware	Collection	Features / format	Size	Max count	Missing counts / imbalance	Still vs moving
12	Haron98 WIFI-CSI-Dataset	3 ESP32 receivers (model not stated; tagged esp32) logging one router/AP (MAC 50:ff:20:e5:a8:f3). Antenna index 0	ESP32 CSI_DATA logging of router traffic. Channel 6, bandwidth flag 0 (20 MHz), MCS 0-7, ~23-28 Hz (0.039-0.044 s between frames), ~4 s per trial	.data text logs: PC timestamp + 24 CSI_DATA fields (seq, MAC, RSSI, rate, sig_mode, MCS, noise floor, channel, local timestamp, ...) + 128 ints (64 complex subcarriers, I/Q)	4,800 files = 1,600 trials x 3 receivers (verified on the repo: 300 files per person per label, 228.8 MB). 480,000 frames, 479,402 valid	Could not verify	**Not a counting dataset as documented**: 4 classes label_00-03 with no names. Balanced: 4 subjects x 4 labels x 100 trials	Not specified
13	OTH Regensburg thesis set	2x ESP32-S3-DevKitM-1 (TX + RX), Raspberry Pi 5 logger. Antennas not specified	Espressif ESP-CSI firmware (ESP-IDF v6), ESP-NOW, 2.4 GHz ch11, 30 pkt/s nominal (24-29 Hz measured in samples). Firmware source is set to HT40	CSV: PC timestamp, label, room tag, free-text note, raw ESP-CSI line (384 ints = 192 subcarrier entries, RSSI)	~120 sessions claimed, ~200,000 packets. **Public: 4 sample files (one ~60 s file per class)**; full set on request	3	0-3 claimed. Sessions per count: Not specified (only the 4 public samples verified, 1 per class)	Both: notes show sitting, standing and slow walking

Presence datasets (empty vs one person), plus RF_ESP32

These support only 0 vs 1 person (RF_ESP32 adds an un-numbered "multiple people" class). All use ESP32 receivers.

#	Dataset	Hardware	Collection	Features / format	Size	Max count	Missing counts / imbalance	Still vs moving
14	ESPectre (data/ folder)	ESP32, ESP32-S3, -C3, -C5 and -C6 receivers on a home Wi-Fi AP. AP model not specified	ESPectre firmware, HT20 / HT-LTF, nominal 100 pkt/s (95-122 measured)	.npz per recording: int8 CSI (N x 128 = 64 complex subcarriers), RSSI, noise floor, channel, PHY mode, device and Wi-Fi timestamps	120 recordings, ~337 min, ~2.13 M packets, ~181 MB; 4 environments (bedroom, living room, hobby room, vacation home)	Not specified ("human presence")	Empty 20 files (112 min); static presence 50 (150 min); motion 50 (75 min). Count is not labelled	Both: separate static-presence and motion recordings
15	esp32-csi-pipeline	2x ESP32 DevKit (TX + RX), ~10 ft apart (26 ft in room 2)	ESP-IDF CSI; receiver picks the quietest 2.4 GHz channel (or the busiest, in the noisy set). ~100 Hz assumed, no timestamps	CSV: packet counter + amplitude of 64 subcarriers. No phase, RSSI or timestamps	9 recordings (3 room/channel setups x 3 states), ~239,500 rows, ~42 MB	1	0: 3 recordings; 1: 6 (3 stationary, 3 moving). Single subject	Both: "stationary" = seated, breathing; "moving" = walking
16	HomeOccupancy	2x ESP32-C6 (TX in AP mode, RX in STA mode), ~3 m LOS	802.11n HT-LTF, 20 MHz, channel 6 in data, ~200 pkt/s (irregular)	CSV per packet: 128 ints (64 complex subcarriers), RSSI, noise floor, AGC/FFT gain, local timestamp	9 recordings, ~1.7 M packets, ~720-754 MB; one home room	1 (implied by labels, not stated)	0: 3 recordings (empty); 1: 6 (3 sleep, 3 work)	Both: sleep = lying still; work = desk typing (subtle motion)
17	OpenCSI	ESP32 meshes (ESP32-S3, a second S3 batch, C3, C6); 8 or 10 nodes per manifest	UDP CSI to a backend, 20 Hz nominal per session. S3/C3: 802.11n HT20, 52 subcarriers; C6: 802.11ax HE20, 246 subcarriers, 2.4 GHz	Raw CSI per link, pre-computed Z-scores, calibration state, annotations.jsonl with timestamped labels	4 sessions x 12 min, 3 rooms, ~691-724 MB	1	Per session: empty 5 + 2 min, static 3 min, moving 2 min. Single subject (S01)	Both: 3 min seated/standing, 2 min walking/gesturing
18	Wallhack1.8k	ESP32-S3 with biquad (BQ) or built-in PIFA antenna	Espressif esp-csi, 100 Hz. Band and bandwidth not specified	CSV per recording: raw CSI (256 ints), RSSI, rate, bandwidth, local timestamp; 1,806 derived spectrograms	44 recordings (~737 k packets), 135 MB zip; LOS hallway and NLOS through 4 brick walls	1	0: 4 recordings (one per scenario/antenna); 1: 40 (20 walking, 20 walking + waving). Subject count not stated	Moving only (no still class)
19	RF_ESP32_Dataset	TX: TP-Link AP, 3 antennas. RX: ESP32 (esp-csi), 1 antenna	2.4 GHz, 20 MHz per README, 64 subcarriers	Per session: csi_data.npy, agc/fft .npy, events.csv, metadata.json (object type, activity, environment, distance)	192 sessions (~132 MB); human-activity subset: 71 sessions, ~2.58 M frames, ~31 MB	"multiple_people" (number not given)	Human-activity sessions: no object 4, one person 27, multiple people 39, mixed 1. No numeric counts	Both: sitting, standing, stationary, walking, running, rotating

Per-dataset notes

1. RadioPoints

- **[Note]** The README says 0-7, but the repo also contains 8p files in rooms A and C.
- Each count is one continuous file (0p...8p) per room, not multiple sessions. The README doesn't say whether people moved.
- The repo is GPL-3.0, but the README separately says the data is for non-commercial research.

2. WiFlow

- **[Note]** The source note says "exact, timestamped headcounts", but the paper labels enter/exit/no-event events, timed by beeps.

- Recorded in a lab (7x7 m) and a classroom (5.5x9 m) with 10 participants. Recording, session and per-count numbers: Not specified. It is not publicly downloadable.

3. EHUNAM

- Counts come from the official Summary.xlsx. The number of files is larger than the number of captures because every capture is saved once per receiver (3-4 receivers).

Count	0	1	2	3	4	5	6	7	8
Distinct captures (PC; count 0 = the empty "E" captures)	123	70	69	65	39	10	10	10	**2**
Extra captures from industrial-lab recordings with machines running (PC+MAR)	-	38	11	6	2	2	-	-	-

- Counts 5-8 exist only in the classroom and Office 1. The industrial lab (MC3) covers 1-3 people only.
- **[Note]** The "21 people" figure is the number of participants, not an occupancy count.
- Movement: the paper says people walked freely, and in the 4- and 8-person sessions one person left each minute. No still-occupancy counting recordings are documented.

4. 80 MHz Human Sensing

- **[Note]** The source note says 1-10 people. The counting folders hold n00 and n02-n10, so 1 person is missing (it comes from the PI folders).
- The paper says people were "moving in the environment" during counting traces. Still activities (sitting still, standing still) appear only in the single-person activity-recognition folders, not in the counting folders.

5. WiCount

- Recording counts and packet totals come from a local audit of the downloaded files.
- One room, one day (2023-12-21). Measured frames are all legacy non-HT (sig_mode=0), although the README says 802.11n. The licence is not declared.

6. WiMANS

- This is an activity-recognition dataset. The count labels are derived from the number of annotated users, not collected as a separate counting set.
- Per-user activities: nothing, walking, rotation, jumping, waving, lying down, picking up, sitting down, standing up. Still occupancy appears only as "nothing" or "lying down" activity labels. Three environments: classroom, meeting room and empty room.

7. CountFi

- Recorded on the upper deck of a Hong Kong double-decker bus. The paper says passengers did "static activities (sitting, talking, sleeping)".
- The 2025 add-on sets are labelled _stop (stationary) and _xm (mixed moving and stationary).
- The paper's 790 segments per class are windows cut from recordings, not independent recordings. Licence: CC BY-NC 4.0.

8. WiFiVision-Counting-Tools

- **[Note]** The overview says "7+ people", but the class table ends at "Seven people"; class 7 may include more than 7.
- One lab room (5x4 m). The page mentions "standardized people positioning and movement patterns" but doesn't say whether people sat, stood or walked.
- Repo code is Apache-2.0; the dataset is academic-use only, by Google Drive access request.

9. OfficeLocalization

- This is a zone-localization dataset with at most one person. Its labels are locations, not people counts.
- **[Note]** The card lists 8 files (2 per class), but the repo now holds 11: five empty recordings and two per zone.
- Recorded October 2025 in an office. Licence: CC BY 4.0.

10. Less is More mesh

- The diary (daily_annotations.yaml) records 2 occupants for Jan 1-11 and 1 from Jan 12, but the hourly labels are states (empty / sleep / active), not counts.
- Labels come from one resident's recall in 2-hour blocks. Camera metadata (no images) covers only ~33 h. The authors warn that labels follow a strong time-of-day pattern.
- The README mentions a separate office deployment (13 nodes, 27 sessions) with camera person counts. That is a different dataset and is not in this record. Licence: CC BY 4.0.

11. IIT (BHU) multi-human

- Activity-recognition dataset; user counts are scenario groups (1, 2 and 3 users), not a separate counting set. More than 80 participants, six activities from subtle hand movements to full-body actions; activity names are not given on the page.
- Files require an IEEE DataPort subscription. DOI 10.21227/e6c6-aa21.

12. Haron98 WIFI-CSI-Dataset

- The card names the dataset "WiFi CSI Router Reception Dataset". Label names are left as TODO, and nothing states that labels are people counts. Each trial belongs to one subject (id_person_01-04), and every trial is recorded by all 3 receivers at once, so the 3 files of a trial are not independent recordings.
- All data was recorded on one evening (2026-04-18, 20:17-23:21, UTC+3). No train/test split is provided.
- Data quality: 598 of 480,000 frames (0.125%) are malformed, always line 1 of a file; 598 of 4,800 files are affected.
- **[Note]** The card describes a wifi_data_set/ folder, a raw dash-encoded layout and restructure_dataset.py/.ps1 scripts, but the repo holds only id_person_01-04 at the root plus README.md (checked through the Hugging Face API).
- **[Note]** The YAML header says MIT, but the card body says no licence is included.

13. OTH Regensburg thesis set

- Master's thesis, OTH Regensburg, 2026. Rooms: bedroom (42 sessions), kitchen (30), classroom (26). Four subjects, recorded over multiple days.
- **[Note]** The room table sums to 98 sessions, not the ~120 stated.
- Posture: the notes on the 4 public samples read "empty","1p sitting", "2p both sitting" and "3p all sitting". The repo's partial campaign log (12 files, one day, counts 0-1 only) adds "1p standing" and "1p walking slow". The full set's mix is not documented.
- No licence on the data and no contact address in the README; the firmware files carry Espressif's Apache-2.0 header. Full set by request through the GitHub account.

14. ESPectre

- Labels are empty / static presence / motion. The project README says it "does not identify people, count them, or prove that a room is empty", so the number of people in presence recordings is not recorded.
- All files come from one contributor. The repo's own quality check gives 6 static/motion pairs a warning or fail. Licence: GPL-3.0.

15. esp32-csi-pipeline

- Undergraduate project. One room recorded on its quietest channel and on the most congested channel, plus an unseen second room. Amplitude only.
- The README notes: no real timestamps (100 Hz assumed), single test subject. Data CC BY 4.0, code MIT.

16. HomeOccupancy

- Same authors and hardware as OfficeLocalization (row 9). Files _1/_2 are train, _3 test. Licence: CC BY 4.0.

17. OpenCSI

- **[Note]** The datasheet says 8 nodes and 56 links per session, but the manifest assigns envB a 10-node topology.
- Every session follows one script: 5 min empty baseline, 3 min static, 2 min moving, 2 min empty. Licence: CC BY 4.0.
- **[Note]** The datasheet says 20 Hz, but an earlier check in this project (V3) found per-link data arriving in bursts at about 1 snapshot/s.

18. Wallhack1.8k

- Built for long-range through-wall activity recognition. Every occupied recording is walking, so there is no still-occupancy class.
- **[Note]** Zenodo lists CC BY 4.0, but the page also says non-commercial research only.

19. RF_ESP32_Dataset

- Only the 71 human-activity sessions carry person labels; the other 121 sessions are metal objects and environments. Labels are attached to events (200-packet snippets) within a session.
- "multiple_people" never records how many people. Data CC BY 4.0, tools Apache-2.0.